# Real-Time Linux Networking and Troubleshooting Commands

### 🎯 Objective:

To gain hands-on experience using Linux commands for network configuration, troubleshooting, and system monitoring by simulating real-world industry scenarios.

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### 🏭 Industry Scenario:

You are working as a **Junior DevOps Engineer** at a **SaaS company** called **TechCloud Inc.** One of your production web servers (Ubuntu-based) is experiencing **intermittent connectivity issues and slow response times.**

Your **task** is to investigate the issue using essential Linux commands, document your findings, and report actionable insights.

### ✅ Prerequisites:

- Running EC2 instance (Ubuntu)
- Create IAM user
    - Give EC2 full permissions
    - Access EC2 via IAM user

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# 🧪 Task Instructions

### 🔹 Task 1: Check IP and Connectivity

1. **Command:** `ip a`

    - **Objective:** List all network interfaces and their IPs.
        i. Displays all IP addresses and network interfaces on the system.
        ii. Shows your computer's internet connection details.

**Expected Output Snippet:**

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500
    inet 192.168.0.101/24 brd 192.168.0.255 scope global dynamic
ens33
```

2. **Command:** `ping google.com`
    - **Objective:** Test internet connectivity.
        i. Tests network connectivity to Google by sending ICMP echo requests.
            1. ICMP (Internet Message Control Protocol) is protocol that helps troubleshoot and manage network problems
        ii. Checks if your computer can reach Google.

**Expected Output:**

```
PING google.com (142.250.190.14) 56(84) bytes of data.
64 bytes from lax02s29-in-f14.1e100.net: icmp_seq=1 ttl=115
time=18.5 ms
```

3. **Command:** `curl ifconfig.me`
    - **Objective:** Get the server's public IP.
        i. Retrieves your public IP address using an external service.
        ii. Tells you your internet (public) IP address.

**Expected Output:**

```
203.0.113.45
```

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### 🔹 Task 2: Check Open Ports and Services

4. **Command:** `netstat -tulnp`
    - **Objective:** List all open ports and running services.
        i. Shows all listening ports and associated processes (TCP/UDP).
        ii. Lists which programs are using the internet or network.

**Expected Output:**

```
tcp        0      0 0.0.0.0:80        0.0.0.0:*     LISTEN
1234/nginx: master
```

5. **Command:** `ss -tuln`
    - **Objective:** Use modern replacement of netstat.
        i. Displays listening TCP/UDP sockets without resolving names.
        ii. Shows which apps are ready to connect to the internet.

**Expected Output:**

```
LISTEN 0      128       *:22                    *:*
LISTEN 0      128       *:80                    *:*
```

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### 🔹 Task 3: DNS and Routing

6. **Command:** `wget htcleaom/sample.txt`
    - **Objective:** Test web download functionality.
        i. Downloads a file from the specified URL.

1. https://www.filesampleshub.com/format/document/txt#google_vignette
   ii. Downloads a file from a website.

**Expected Output:**

```
Saving to: ‘sample.txt’
sample.txt             100%[====================>]   127
--.-KB/s    in 0s
```

7. **Command:** `traceroute google.com`
   - **Objective:** Diagnose routing path to Google.
     i. Traces the path packets take to reach Google.
     ii. Shows the path your internet traffic takes to reach Google.

**Expected Output:**

```
1  192.168.0.1  1.123 ms
2  10.10.0.1    3.456 ms
3  142.250.190.14  25.678 ms
```

8. **Command:** `ifconfig`
   - **Objective:** (Deprecated) Still check interfaces (alternative to `ip a`).
     i. Shows network interfaces and IP configuration (older tool)
     ii. Shows internet and network settings (older method).

**Expected Output:**

```
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
```

```
      inet 192.168.0.101  netmask 255.255.255.0  broadcast
192.168.0.255
```

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9. **Command:** `nslookup google.com`

    - **Objective:** Check DNS resolution.
        i. Queries DNS to resolve Google's IP address.
        ii. Finds the IP address of a website.

**Expected Output:**

```
Name:    google.com
Address: 142.250.190.14
```

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### 🔹 Task 4: Troubleshoot and Audit Logs

10. **Command:** `grep "error" /var/log/syslog`

    - **Objective:** Search for error logs.
        i. Looks for errors in the system log.

**Expected Output:**

```
May 12 07:45:10 server1 nginx[1234]: [error] 500 Internal Server
Error
```

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11. **Command:** `find /etc -name "nginx*"`

    - **Objective:** Locate nginx-related configuration files.
        i. Finds all files in `/etc` matching the name pattern "nginx*".

**Expected Output:**

```
/etc/nginx
/etc/nginx/nginx.conf
```

12. **Command:** `history`
    - **Objective:** View previously executed commands.

**Expected Output:**

```
101  ip a
102  ping google.com
103  grep "error" /var/log/syslog
```

13. **Command:** `cut -d':' -f1 /etc/passwd`
    - **Objective:** List all usernames on the system.

**Expected Output:**

```
root
daemon
syslog
Ubuntu
```

### Task 5: Configure UFW Firewall

A **firewall** is like a **security guard** for your computer or server. It decides what is allowed in and what is blocked based on rules.

On a web server, we allow:

✅ Port **22** (SSH) for admins to manage the server
✅ Port **80** (HTTP) for people to view the website
❌ Block all other ports to protect from hackers

**[Commands to Execute]**

```
sudo apt install ufw              # Install if not present
sudo ufw enable                   # Enable firewall
sudo ufw allow ssh                # Allow SSH port (22)
sudo ufw allow http               # Allow HTTP port (80)

          # View rules

sudo ufw status verbose
```

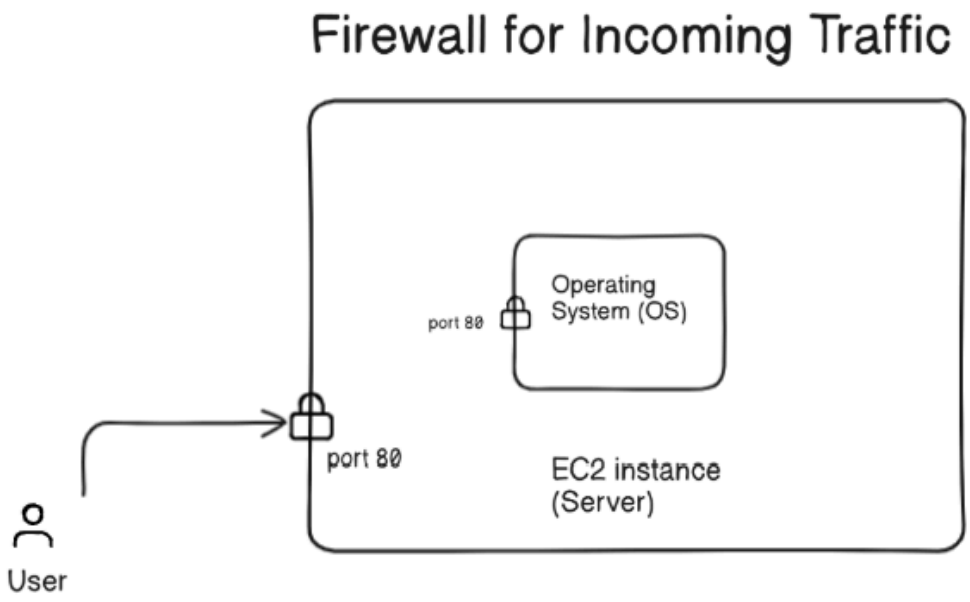
**Expected Output:**

```
Status: active

To                         Action      From
--                         ------      ----
22                         ALLOW       Anywhere
80                         ALLOW       Anywhere
```

**Verify the firewall**

```
sudo netstat -tuln | grep :80
```




## Conclusion / Summary

This class gave you hands-on practice with essential Linux commands used daily by DevOps and system admins. You simulated a real-world scenario to troubleshoot network issues, check system logs, and verify connectivity and DNS.

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### 🎓 Key Takeaways:

- ✅Learned to use core networking and troubleshooting commands
- ✅Practiced diagnosing real system issues
- ✅Improved command-line and log analysis skills
- ✅Secure a server using the UFW firewall an security groups

You're now better prepared to handle real production environments. Great job!